

HW4 PROBLEM LOOKUP — READ FIRST

#	Question → section
P1	Power density of a dipole → DIPOLE TYPE then HERTZIAN
P2	HPBW β , solid angle Ω_p , directivity D from $F(\theta)$ → BEAM PARAMETERS
P3	$\theta_{\max}$, $S_{\max}$, plot $F(\theta)$ for $\lambda/4$ dipole → ARBITRARY-LENGTH DIPOLE
P4	Effective area A_e , ratio A_e/A_p → EFFECTIVE AREA
P5	Friis link budget (P_r given P_t, G_t, G_r, R) → FRIIS FORMULA

Step 0 always: $\lambda = c/f$. Step 1 for any dipole: classify by l/λ .

DIPOLE TYPE BY LENGTH

Compute first $\lambda = c/f$, then ratio l/λ		
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — you'll get the wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$ (P1)

Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32 \pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$
$k = 2\pi/\lambda$ (rad/m); I_0 peak current (A); l length (m); R obs. dist (m); $\eta_0 = 120\pi \approx 377 \Omega$.
Max direction broadside $\theta_{\max} = \pi/2$
Recipe 1. Compute $\lambda = c/f$. Check $l/\lambda < 1/50$.
2. Compute $k = 2\pi/\lambda$.
3. Plug into $S(R, \theta)$ with given R, θ .
HW4 P1: $l = 1\text{m}$, $f = 1\text{MHz}$, $I_0 = 12\text{A}$, $R = 5\text{km}$, $\theta = 45^\circ$. $\lambda = 300\text{m}$, $l/\lambda = 1/300 < 1/50 \checkmark$. $k = 2\pi/300$.
$S = \frac{(120\pi)(2\pi/300)^2(144)(1)}{32\pi^2(5 \times 10^3)^2} \cdot \frac{1}{2} = \mathbf{1.51 \times 10^{-9} \text{ W/m}^2}$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$ (P3)

Power density at R any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2} \right)^2 \approx 0.0858 S_0$
Normalized pattern plot this for P3c
$F(\theta) = \frac{S(\theta)}{S_{\max}} = \frac{1}{0.0858} \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$
HW4 P3: $\theta_{\max} = \pi/2$ (broadside); $S_{\max} = 0.0858 S_0$; $F(\theta)$ peaks = 1 at $\pi/2$, drops to 0 at $\theta = 0, \pi$.
Half-wave $l = \lambda/2$ (ref. only) $\pi l/\lambda = \pi/2$
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$

BEAM PARAMETERS — β, Ω_p, D (P2)

Step 1 — Normalize <i>always start here</i>
$F(\theta) = S(\theta)/S_{\max}$
Step 2 — HPBW β solve $F(\theta_{1/2}) = 0.5$, then $\beta = 2\theta_{1/2}$ for symmetric beams
Gaussian shortcut: $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$.
HW4 P2a: $F = e^{-20\theta^2}$, $a = 20 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/20} = 0.186$ rad. $\beta = 0.372 \text{ rad} = 21.31^\circ$.
Step 3 — Pattern solid angle Ω_p units: sr
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.
Most integrals have NO closed form. Use Wolfram / numerical.
Wolfram: <code>2*pi*Integrate[Exp[-20 t^2]*Sin[t],{t,0,pi}]</code>
HW4 P2b: $\Omega_p = 0.156 \text{ sr}$ (numeric).
Step 4 — Directivity D
$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$
HW4 P2c: $D = 4\pi/0.156 = \mathbf{80.55}$ (19.06 dB).
Step 5 — Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73

Half-wave is the default “dipole antenna” unless stated otherwise.

EFFECTIVE AREA A_e (P4)

Definition <i>antenna's RX cross section (m²)</i>
$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$
Physical cross section <i>wire dipole, diameter d</i>
$A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$
Recipe 1. $\lambda = c/f$. 2. Pick D (1.64 for half-wave).
3. $A_e = \lambda^2 D/(4\pi)$. 4. $A_p = l d$. 5. Ratio.
HW4 P4: half-wave, $f = 100\text{MHz}$, $d = 2\text{cm}$. $\lambda = 3\text{m}$, $D = 1.64$. $A_e = \frac{9(1.64)}{4\pi} = \mathbf{1.17 \text{ m}^2}$. $A_p = (1.5)(0.02) = 0.03 \text{ m}^2$. Ratio $\mathbf{A_e/A_p = 39.2}$.
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

FRIIS TRANSMISSION FORMULA (P5)

Far-field link budget P_t tx, P_r rx, range R
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$
G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R max range
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$
Equivalent form using A_e <i>recv. area form</i>
$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
Recipe 1. $\lambda = c/f$.
2. Convert any dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.
3. If TX is “half-wave dipole” use $G_t = 1.64$.
4. Plug into Friis.
Use linear G , not dB. Convert FIRST.
HW4 P5: half-wave TX, $P_t = 1\text{kW}$, $f = 50\text{MHz}$, RX $G_r = 3\text{dB} = 2$, $R = 30\text{km}$. $\lambda = 6\text{m}$, $G_t = 1.64$. $P_r = (1.64)(2) \left(\frac{6}{4\pi \times 10^4} \right)^2 10^3 = \mathbf{8.3 \times 10^{-7} \text{ W}} = 0.83 \mu\text{W}$.

dB ↔ LINEAR

$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$			
dB linear		dB linear	
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

SYMBOL REFERENCE

l — antenna length (m); $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
I_0 — peak current (A); $\eta_0 = 120\pi \approx 377 \Omega$
$k = 2\pi/\lambda$ wavenumber (rad/m)
R — range from antenna to obs. pt (m)
$S(R, \theta)$ — power density (W/m ²)
$F(\theta) = S/S_{\max}$ — normalized pattern (unitless)
β — half-power beamwidth (rad or deg)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity (unitless)
ξ — radiation efficiency ($0 < \xi \leq 1$); $G = \xi D$
$A_e = \lambda^2 D/(4\pi)$ — effective area (m ²)
$A_p = l d$ — physical cross sec. (m ²)
P_t, P_r — TX / RX power (W); G_t, G_r linear gains